

Motivation

- Portfolio optimization [1] requires dynamic allocation of funds by balancing the risk and return trade-off under dynamic market conditions.
- Deep Reinforcement Learning (DRL) has emerged as a promising tool for adaptive portfolio management, learning optimal asset allocation through interaction with market environments and capturing complex, nonlinear patterns.
- However, these models trained on historical data often under-perform in crises due to distribution shifts, rare extreme events, and struggle to fully capture the complex market dynamics.
- Generative models have gained significant traction in finance, particularly for scenario generation and stress testing. These synthetic scenarios are useful both for portfolio stress testing and enrich training datasets for more robust machine learning models in finance [2].
- Diffusion models, particularly Denoising Diffusion Probabilistic Models (DDPMs) [3], excel at generating realistic samples by reversing a noise-addition process and can be conditioned for targeted scenarios like market crashes.

DRL Framework for Portfolio Optimization

- The portfolio optimization problem is formulated as a Markov Decision Process (MDP).
- The uncertainty of future prices transforms it into a stochastic control problem, which can be solved using model-free DRL methods.
- Stock prices represent the state and satisfy the Markov property.
- A state of the environment consists of the covariance matrix of close prices and some commonly used technical indicators.
- At each time step t , for each stock, the agent can select an action $w_t \in [w_1, w_2, \dots, w_n]$, which denotes the amount of funds to be allocated in that particular stock, respectively.
- The goal of the agent is to learn a policy $\pi(a_t|s_t)$ that maximizes the expected cumulative return by interacting with the environment at discrete time steps [4].

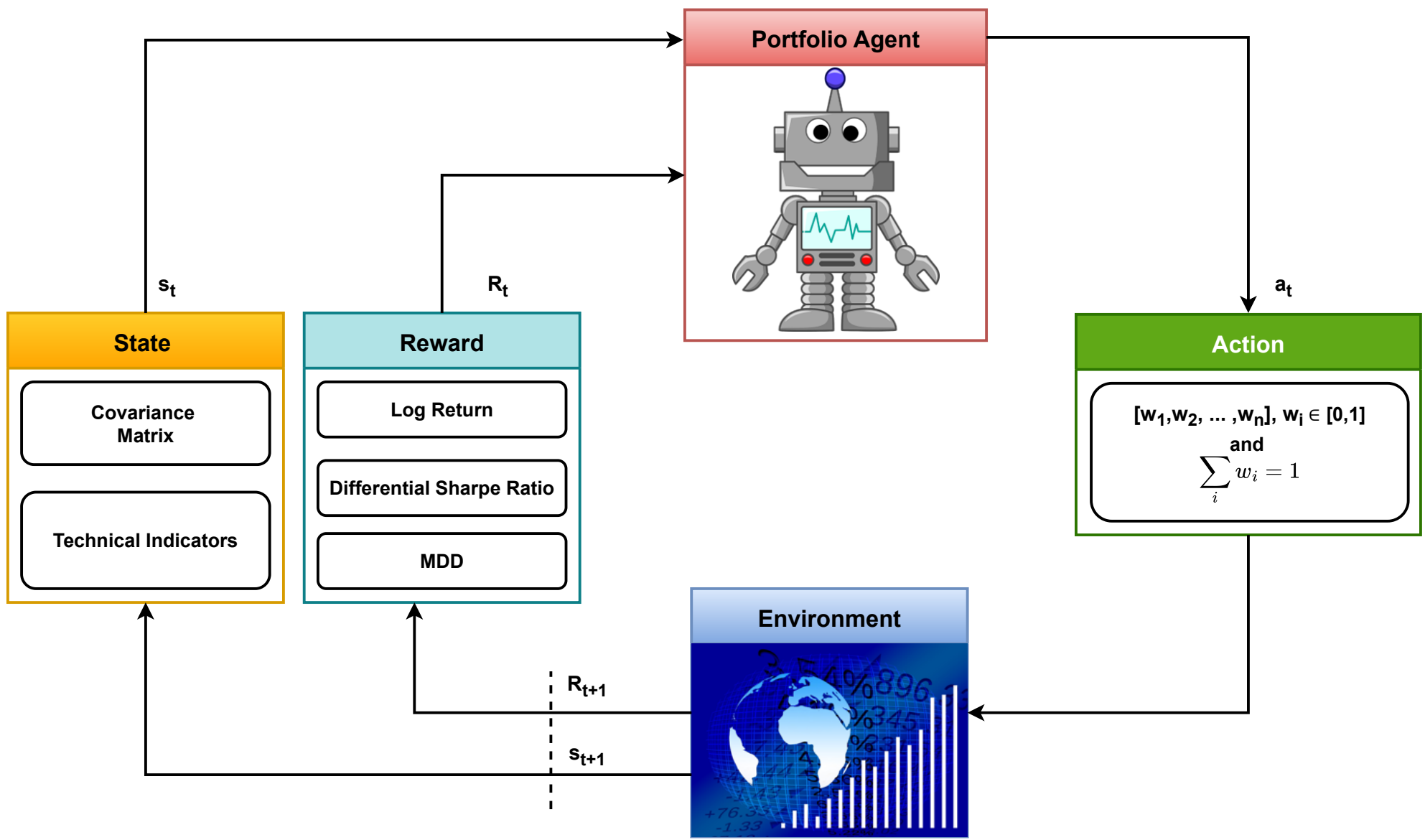


Figure 1. Overview of the portfolio optimization environment using Deep Reinforcement Learning.

Proposed Methodology

- This study proposes an innovative framework, termed Diffusion-Augmented Reinforcement Learning (DARL), which integrates DDPMs with DRL for robust portfolio management.
- Diffusion models can generate diverse, realistic samples conditioned on specific attributes (e.g., crash intensity), simulating tail-risk scenarios absent or underrepresented in historical data.
- Augmentation expands the training dataset without collecting new real data, mitigating data scarcity in finance where crises occur infrequently.
- This leads to better policy generalization, by encouraging the agent to learn invariant features across augmented and real data.
- Augmentation integrates seamlessly into DRL by appending synthetic episodes to the replay buffer or environment, enabling the agent to explore high-reward policies in simulated stress.
- Train a Proximal Policy Optimization (PPO) [5] agent to optimize portfolio weights.

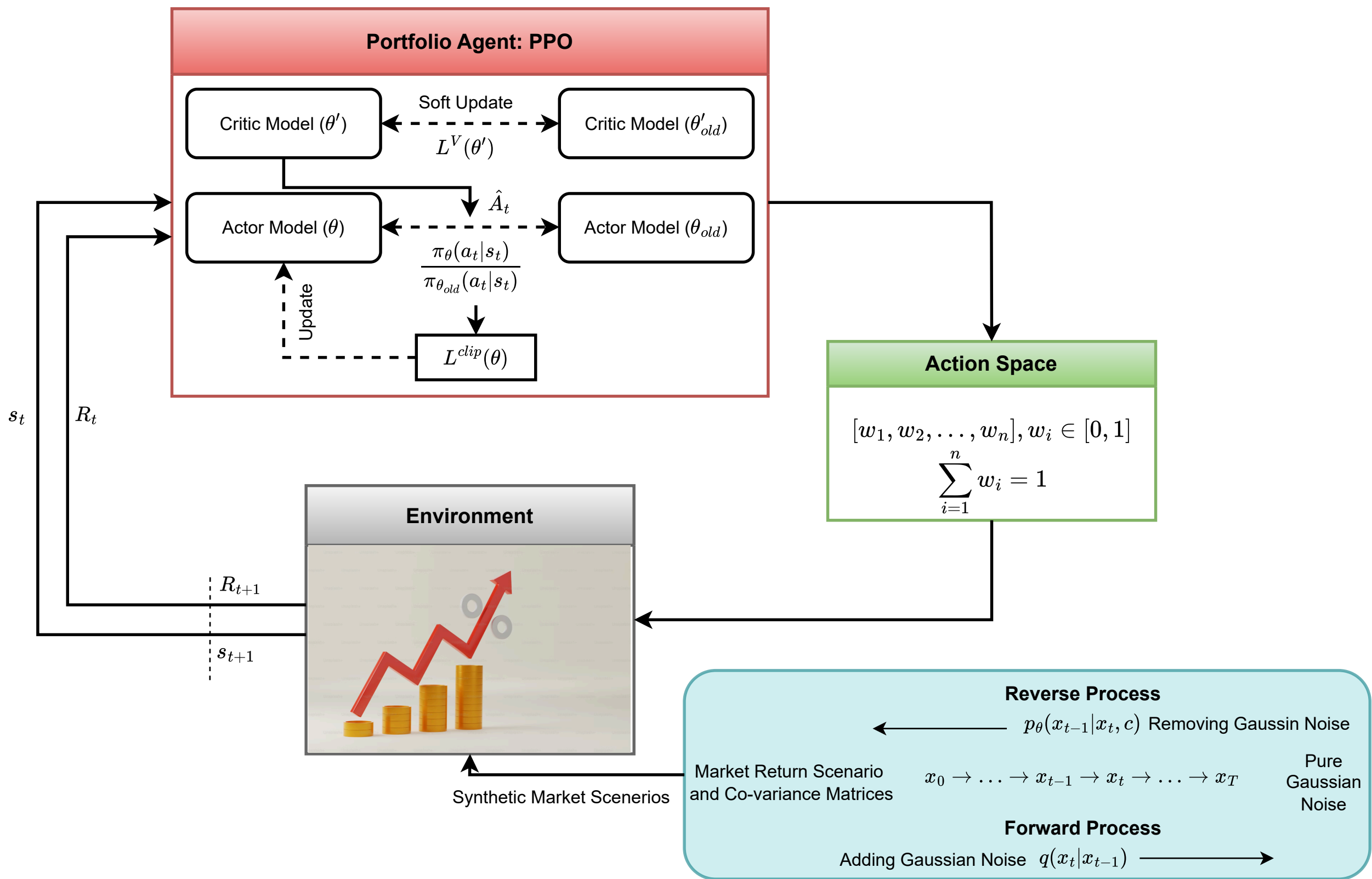


Figure 2. The schematic diagram depicting the workflow of the proposed methodology.

Experiments

- Dataset:** Daily price data for all 30 stocks of the Dow Jones Industrial Average (DJI).
- Train/Test Split:**
 - Train: January 2011 - December 2023
 - Test: January 2024 - July 2025
- Initial Capital:** \$1 Million
- Transaction Cost:** 0.05%
- DRL Tuning:** Bayesian optimization

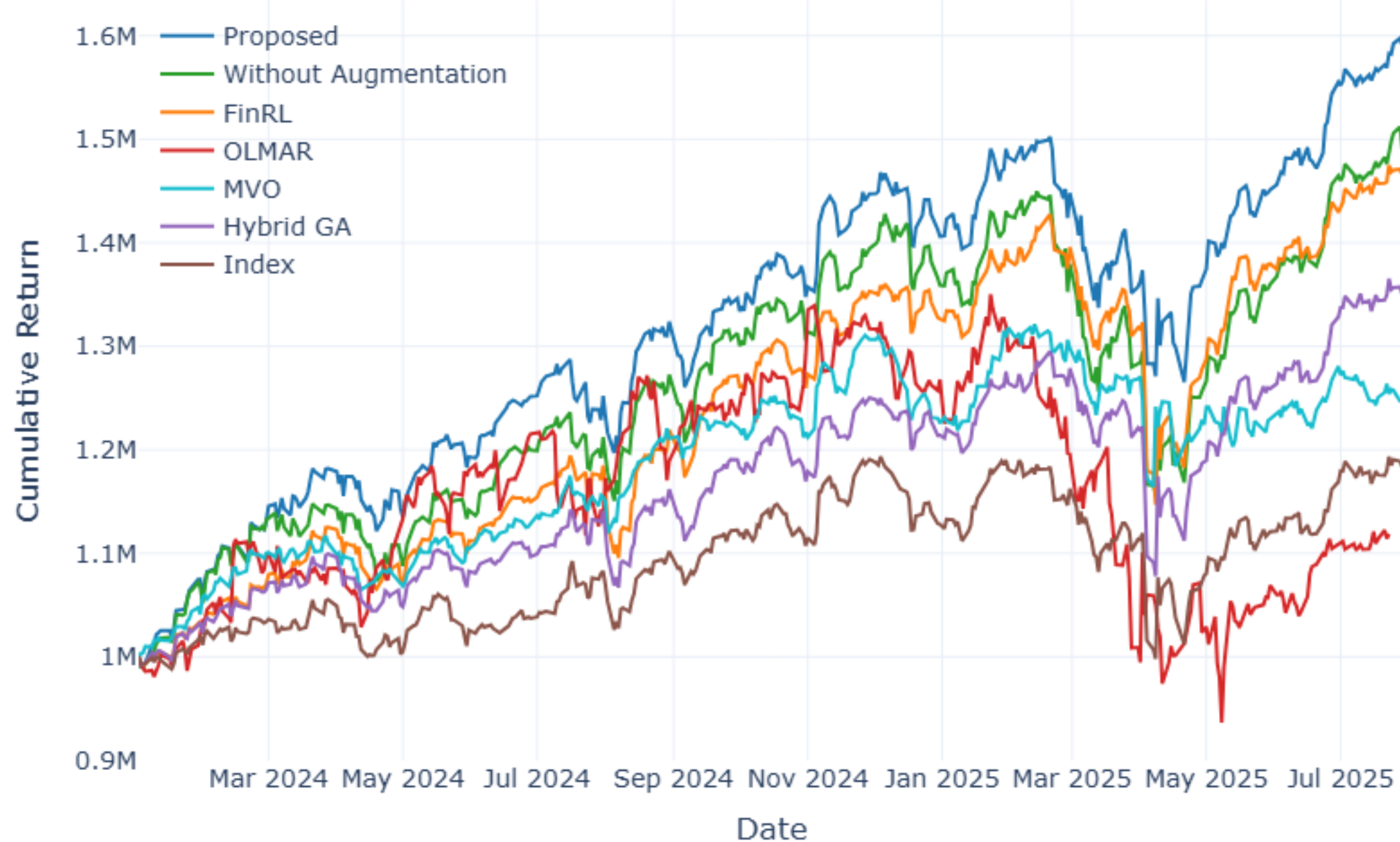


Figure 3. Cumulative return plots of the proposed methodology against the benchmark portfolio strategies over the trading period.

Table 1. Performance evaluation of the proposed methodology against the benchmarks portfolio strategies.

Model/Benchmark	Cumulative Return (%)	Annualized Return (%)	Sharpe Ratio	Calmar Ratio	Annual Volatility (%)	Maximum Drawdown (%)
Proposed	59.5253	34.7101	1.9096	2.2024	16.3058	-15.7598
Without Augmentation	49.4439	29.2149	1.5172	1.4385	17.9649	-20.3080
FinRL-PPO	46.2286	27.4344	1.5411	1.3961	16.6335	-19.6496
OLMAR	11.8773	7.5214	0.4097	0.2455	25.6876	-30.6370
Hybrid-GA	34.5056	20.8184	1.2623	1.2403	15.9922	-16.7852
Markowitz	24.6485	15.1333	1.1246	1.2754	13.3178	-11.8651
Index	17.8874	11.0694	0.7717	0.6762	15.0674	-16.3692

Conclusion

- The Diffusion-Augmented Reinforcement Learning (DARL) framework represents a significant advancement in addressing the challenges of portfolio optimisation in volatile financial markets.
- By integrating DDPMs with DRL, our approach effectively enhances the robustness of portfolio management strategies, particularly during the unseen 2025 Tariff Crisis.
- The use of conditional DDPMs to generate synthetic crash scenarios, parameterised by stress intensity, enriches the training data, enabling the PPO agent to learn policies resilient to tail-risk events.
- Empirical results on Dow 30 stocks demonstrate DARL's superior performance compared to traditional baselines. This methodology offers a practical and scalable solution for investors and asset managers, equipping them to navigate complex market dynamics with greater confidence.
- Future Work**
 - Integrating sentiment analysis from news articles and insights gleaned from company financial reports.
 - Dynamic risk measures like CVaR
 - Include more realistic portfolio constraints during decision making.

References

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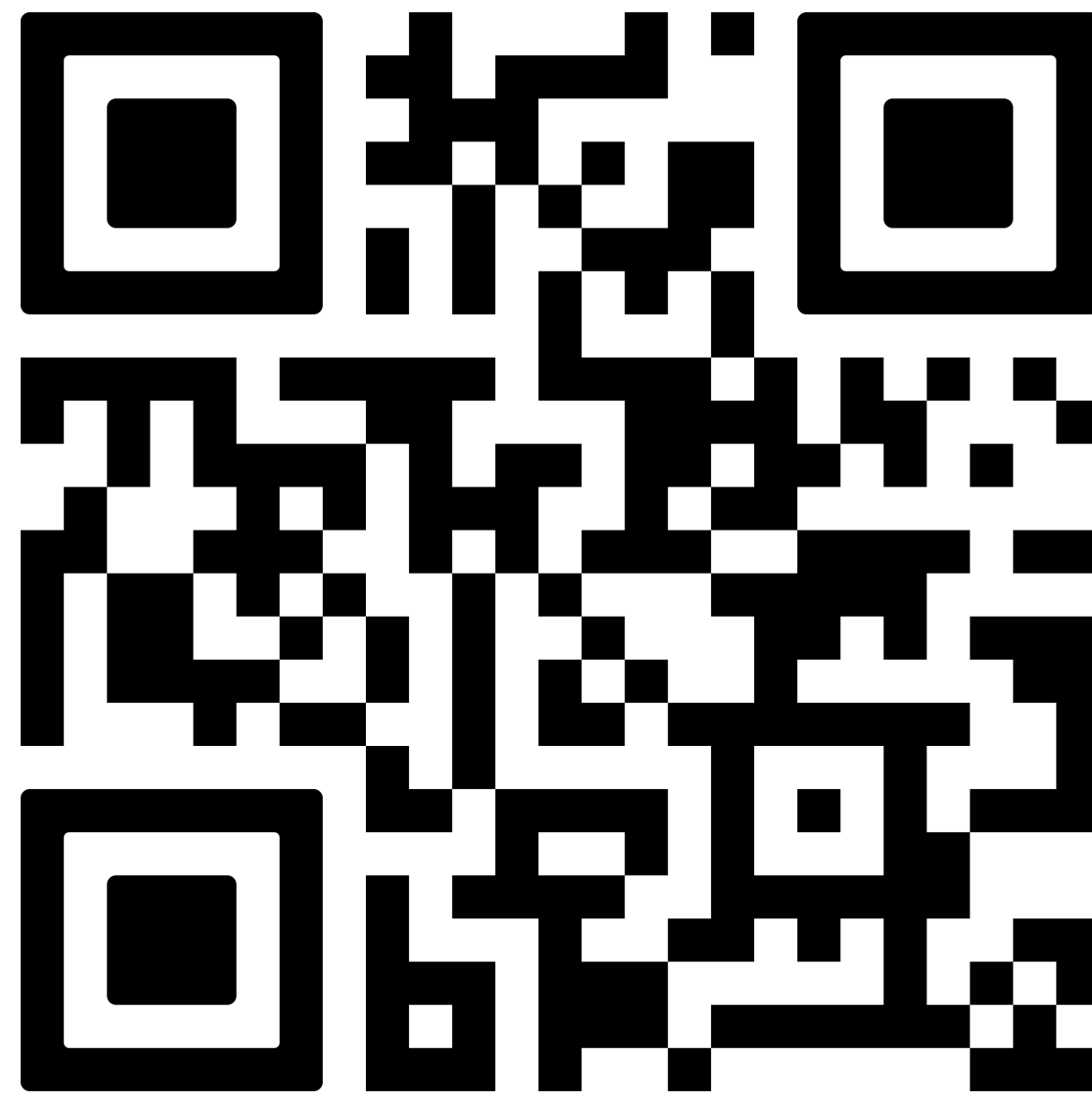


Figure 4. Check out the full paper here.